

Analysis of 2009-2010 Colorado Student Assessment Program Results for Colorado 9th Graders

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Introduction

The Colorado Department of Education (CDE) evaluated Colorado ninth-grade students' proficiency using the Colorado Student Assessment Program (CSAP) in these three subjects: reading, writing, and math. During these years, 2009 -2010, the Colorado math scores were very low, and as a result, Colorado implemented programs to highly focus on the math curriculum in high school. This project analyzes the ninth graders performance from 2009 and 2010 to determine whether their math scores improved and how they relate to their reading and writing results. In specific, we want to address the following questions using the 5A method:

1. Was there a significant improvement in math exam passing rates between 2009 and 2010? If so, was there a decline in reading and writing performance?
2. Is there a relationship between a school's passing rate in math and its passing rates in reading and writing? If so, how accurately can math passing rates be predicted based on reading and writing results?

Ask

The key questions we need to ask ourselves to properly analyze the data for this project are: 1. Did math scores get better from 2009 to 2010? If they did, did reading and writing scores get worse? 2. Do math scores relate to reading and writing scores? If so, how well can we guess math scores using reading and writing scores?

Acquire

To start this study, we need to import the data and load the necessary libraries. The data will come from the Colorado Department of Education (CDE)'s Information Marketplace. The data will include results each of the three subjects: math, reading, and writing. After importing the data for each subject separately, we will combine them into a single data frame for analysis.

```
#importing CSAP data
reading <- read.csv("44v8-6fzj.csv")
writing <- read.csv("44v8-6fzj (1).csv")
math <- read.csv("44v8-6fzj (2).csv")

#combining CSAP data
combine <- rbind(reading, writing, math)
```

Acquire: Clean and Reshape Data

After importing the data, we will clean and wrangle it by filtering out the schools with fewer than 31 students in both 2009 and 2010. Next, we will select the columns needed for our analysis and rename the variables to the appropriate data types for further study. Finally, we will take a random sample of 120 unique schools

and calculate their passing rates by limiting the sample to 120 schools, we will make graphs and plots easier to interpret.

```
## Rows: 1,488
## Columns: 8
## $ year      <dbl> 2009, 2009, 2009, 2010, 2010, 2010, 2009, 2009, 2009, 2~
## $ school_no <int> 10, 10, 10, 10, 10, 10, 15, 15, 15, 15, 15, 15, 24, 24, ~
## $ subject   <fct> READING, WRITING, MATH, READING, WRITING, MATH, READING~
## $ no_score  <int> 1, 0, 0, 2, 3, 4, 0, 0, 0, 0, 0, 0, 17, 16, 18, 9, 9, 1~
## $ unsatisfactory <int> 141, 68, 351, 113, 83, 313, 8, 2, 37, 9, 9, 32, 76, 47, ~
## $ partial   <int> 195, 339, 98, 222, 348, 139, 20, 43, 36, 23, 56, 48, 18~
## $ proficient <int> 151, 81, 34, 174, 78, 51, 66, 44, 20, 86, 50, 31, 148, ~
## $ advanced  <int> 0, 0, 5, 1, 0, 6, 3, 8, 4, 3, 6, 10, 1, 2, 4, 1, 3, 7, ~
```

Acquire: Calculate Passing Rates and Difference of Passing Rates Between 2009 and 2010

In the final step of data acquisition for this study we need to calculate the passing rates. This will help us answer the questions we asked in the beginning of the study. In order to do this we need to obtain the proportion of students who scored “advanced” and “proficient” and divide it by the other possible scores (unsatisfactory, partially proficient, no score). The data was then reshaped to make it easier for the comparison between 2009 and 2010. After that, the difference of passing rates between 2009 and 2010 were calculated for each school. Lastly, the data was split and organized so it could be observed easier.

```
my_schools2 <- my_schools %>%
  pivot_wider(
    names_from = year,
    values_from = c(advanced, proficient, no_score, unsatisfactory, partial)
  ) %>%
  mutate( rate2009 = (advanced_2009 + proficient_2009) /
    (advanced_2009 + proficient_2009 + partial_2009 + unsatisfactory_2009 + no_score_2009),

    rate2010 = (advanced_2010 + proficient_2010) /
    (advanced_2010 + proficient_2010 + partial_2010 + unsatisfactory_2010 + no_score_2010),

    #the passing rate difference of 2010 and 2009
    diff = rate2010 - rate2009

  ) %>%
  #selecting the columns for my_schools2 data frame
  select(school_no, rate2009, rate2010, diff)
#displaying the results in the my_schools2 data frame
glimpse(my_schools2)
```

```
## Rows: 360
## Columns: 4
## $ school_no <int> 15, 15, 15, 40, 40, 40, 65, 65, 65, 118, 118, 118, 203, 203, ~
## $ rate2009  <dbl> 0.71134021, 0.53608247, 0.24742268, 0.35416667, 0.20833333, ~
## $ rate2010  <dbl> 0.73553719, 0.46280992, 0.33884298, 0.36923077, 0.13846154, ~
## $ diff      <dbl> 0.02419698, -0.07327256, 0.09142029, 0.01506410, -0.06987179~
```

Analyze

Analyze: Mean Difference

With the data acquired and cleaned, we go into the data analysis. We will then use bootstrap re-sampling to estimate the sampling distribution of and calculate confidence intervals for the mean difference. First things first in the bootstrap: calculate the statistic of interest. The mean difference between two values in the data frame will be our statistic, to which we will apply repeatedly to analyze the different results from the various re-sampled data frames created during the bootstrap.

```
#calculates the mean difference for bootstrapping
mean_diff <- function(data, indices) {
  #re sampling the data
  resampled_data <- data[indices, ]
  #returning the mean difference
  mean(resampled_data$diff)
}
```

Analyze: Bootstrap Re-sample

We will calculate the mean difference between 2009 and 2010 for each subject. The code will also provide a 95% confidence interval for these differences, and it will visualize the distribution of the differences along with the confidence interval using a plot.

Subject: MATH Bootstrap Mean Difference Estimate: 0.03179104 95% Confidence Interval (Percentile Method): 0.01874932 - 0.04518669

- We are 95% confident that the true mean difference lies within the interval [0.01874932- 0.04518669], since 0 is not included in the interval we can say that there was a positive change in scores.

Subject: READING Bootstrap Mean Difference Estimate: 0.002087836 95% Confidence Interval (Percentile Method): -0.01228427 - 0.01690351

- We are 95% confident that the true mean difference lies within the interval [-0.01228427 - 0.01690351], since 0 is included in the interval we can say that there was not a noticeable change in scores.

Subject: WRITING Bootstrap Mean Difference Estimate: -0.03428596 95% Confidence Interval (Percentile Method): -0.04921054 - -0.01948423.

- We are 95% confident that the true mean difference lies within the interval [-0.04921054 - -0.01948423], since the lowest and highest values of this interval are less than 0, we can say that there was a decrease in scores.

With these 95% confidence intervals, we can see major changes in math and writing.

Analyze: Association Between Subjects

We will now investigate the relationship between the pass rates of the three subjects of math, reading, and writing in the year 2009 with three plots. These plots will show the correlation of the students' performances in each of the subjects.

```
#training a model on the 2009 passing rates data
train_data <- filter(my_schools, year == 2009) %>%
  mutate(total_student=no_score + unsatisfactory + partial +proficient + advanced,
          math_passed = proficient + advanced,
          rate = math_passed/total_student) %>%
  select(school_no, subject, rate) %>%
  pivot_wider(names_from = subject, values_from = rate)
```

```

#displaying the training set
glimpse(train_data)

## Rows: 120
## Columns: 4
## $ school_no <int> 15, 40, 65, 118, 203, 206, 210, 221, 263, 309, 361, 370, 432~
## $ READING    <dbl> 0.7113402, 0.3541667, 0.5227273, 0.4285714, 0.3250000, 0.504~
## $ WRITING    <dbl> 0.5360825, 0.2083333, 0.3409091, 0.3296703, 0.1000000, 0.313~
## $ MATH       <dbl> 0.24742268, 0.05208333, 0.29545455, 0.13736264, 0.00000000, ~

#testing the accuracy of the training set based on the 2010 passing rates data
test_data <- filter(my_schools, year == 2010) %>%
  mutate(total_student=no_score + unsatisfactory + partial +proficient + advanced,
          math_passed = proficient + advanced,
          rate = math_passed/total_student) %>%

  select(school_no, subject, rate) %>%
  pivot_wider(names_from = subject, values_from = rate)

#displaying the testing set
glimpse(test_data)

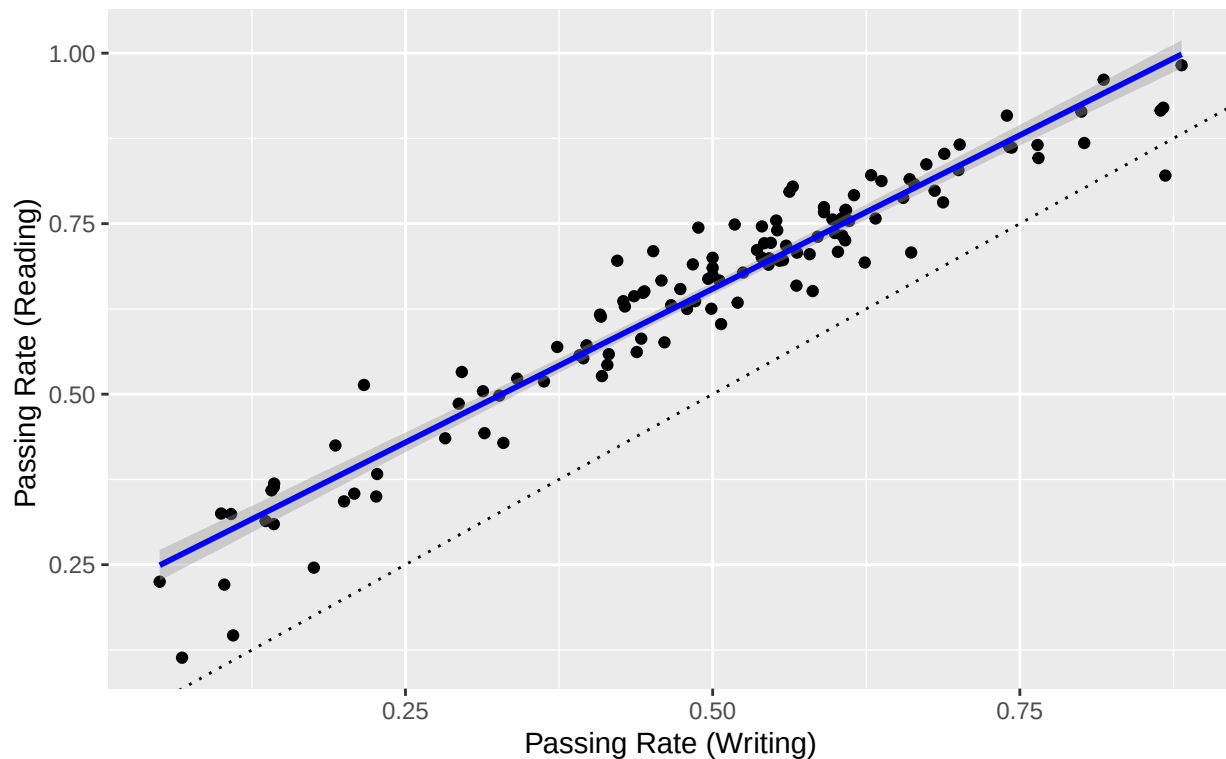
## Rows: 120
## Columns: 4
## $ school_no <int> 15, 40, 65, 118, 203, 206, 210, 221, 263, 309, 361, 370, 432~
## $ READING    <dbl> 0.7355372, 0.3692308, 0.6590909, 0.6369427, 0.2790698, 0.543~
## $ WRITING    <dbl> 0.46280992, 0.13846154, 0.46212121, 0.50318471, 0.11627907, ~
## $ MATH       <dbl> 0.33884298, 0.07692308, 0.38636364, 0.31446541, 0.11627907, ~

#reading & writing plot
plot(ggplot(train_data, aes(x=WRITING, y=READING)) +
  geom_point() +
  geom_smooth(method = "lm", color = 'blue') +
  geom_abline(intercept = 0, slope = 1, linetype = "dotted", color = "black") +
  labs(title = "Colorado Student Assessment Program (CSAP) \n 9th Grade Reading vs Writing (2009)",
        x = "Passing Rate (Writing)",
        y = "Passing Rate (Reading) "))

## `geom_smooth()` using formula = 'y ~ x'

```

Colorado Student Assessment Program (CSAP) 9th Grade Reading vs Writing (2009)



```
#the reading and writing correlation
reading_writing_cor <- cor.test(train_data$READING, train_data$WRITING)
print(paste("Correlation Coefficient:", reading_writing_cor$estimate))
```

```
## [1] "Correlation Coefficient: 0.961425789227356"
```

```
#the linear model for reading and writing data
reading_writing_model <- lm(READING ~ WRITING, data = train_data)
```

```
#summarizing reading and writing model
summary(reading_writing_model)
```

```
##
## Call:
## lm(formula = READING ~ WRITING, data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.165909 -0.024555  0.002288  0.030663  0.114588
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.20416    0.01247   16.37  <2e-16 ***
## WRITING      0.90078    0.02372   37.97  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04983 on 118 degrees of freedom
```

```
## Multiple R-squared:  0.9243, Adjusted R-squared:  0.9237
## F-statistic:  1442 on 1 and 118 DF,  p-value: < 2.2e-16
```

```
#reading and writing predictions & RMSE
```

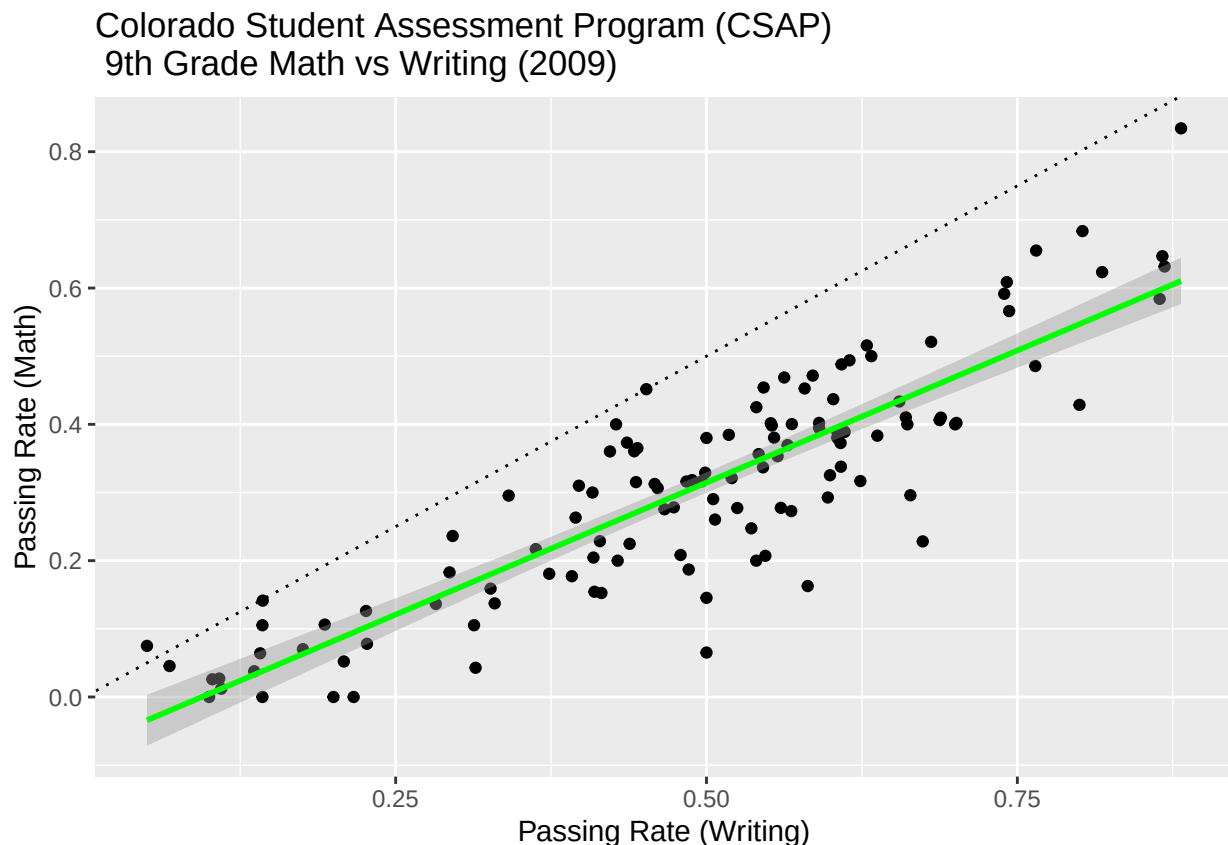
```
reading_writing_prediction <- predict(reading_writing_model,newdata = test_data)
reading_writing_RMSE <- sqrt(mean((reading_writing_prediction - test_data$READING)^2))
print(paste("The RMSE for Reading vs Writing is",reading_writing_RMSE))
```

```
## [1] "The RMSE for Reading vs Writing is 0.0685890265195866"
```

```
#writing & math plot
```

```
plot(ggplot(train_data, aes(x=WRITING, y =MATH)) +
  geom_point() +
  geom_smooth(method = "lm", color = 'green') +
  geom_abline(intercept = 0, slope = 1, linetype = "dotted", color = "black") +
  labs(title = "Colorado Student Assessment Program (CSAP) \n 9th Grade Math vs Writing (2009)", x = "P
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
#linear model for math and writing
```

```
math_writing_model <- lm(MATH ~ WRITING,data = train_data)
```

```
#summarizing math and writing model
```

```
summary(math_writing_model)
```

```
##
```

```
## Call:
```

```
## lm(formula = MATH ~ WRITING, data = train_data)
```

```
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.24936 -0.05151  0.00460  0.06241  0.22396
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.07291    0.02069  -3.524 0.000605 ***
## WRITING      0.77498    0.03935  19.693 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08265 on 118 degrees of freedom
## Multiple R-squared:  0.7667, Adjusted R-squared:  0.7647
## F-statistic: 387.8 on 1 and 118 DF,  p-value: < 2.2e-16

#the math and writing correlation
math_writing_cor <- cor.test(train_data$MATH,train_data$WRITING)
print(paste("Correlation Coefficient:", math_writing_cor$estimate))

## [1] "Correlation Coefficient: 0.87562132538972"

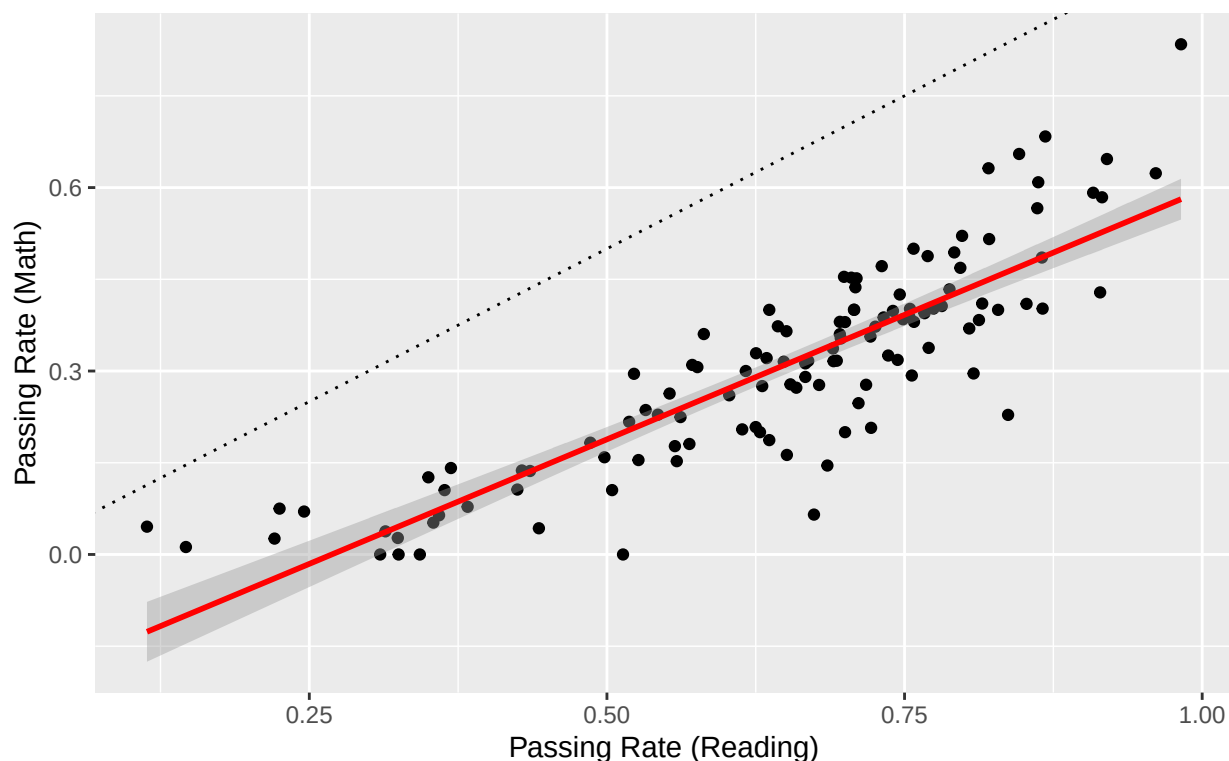
#math and writing predictions & RMSE
math_writing_predictions <- predict(math_writing_model, newdata = test_data)
math_writing_RMSE <- sqrt(mean((math_writing_predictions - test_data$MATH)^2))
print(paste("The RMSE for Math vs Writing is ", math_writing_RMSE))

## [1] "The RMSE for Math vs Writing is  0.095727148022653"

plot(#reading & math plot
ggplot(train_data, aes(x= READING, y =MATH)) +
  geom_point() +
  geom_smooth(method = "lm", color = 'red') +
  geom_abline(intercept = 0, slope = 1, linetype = "dotted", color = "black") +
  labs(title = "Colorado Student Assessment Program (CSAP) \n 9th Grade Math vs Reading (2009)", x = "P

## `geom_smooth()` using formula = 'y ~ x'
```

Colorado Student Assessment Program (CSAP) 9th Grade Math vs Reading (2009)



```
#linear model for math and reading
math_reading_model <- lm(MATH ~ READING, data = train_data)
```

```
#summarizing math and reading model
summary(math_reading_model)
```

```
##
## Call:
## lm(formula = MATH ~ READING, data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.264694 -0.055566  0.000172  0.059665  0.253275
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.21898    0.02950  -7.423 1.94e-11 ***
## READING      0.81448    0.04405  18.489 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08669 on 118 degrees of freedom
## Multiple R-squared:  0.7434, Adjusted R-squared:  0.7412
## F-statistic: 341.8 on 1 and 118 DF, p-value: < 2.2e-16
```

```
#math and reading correlation
math_reading_cor <- cor.test(train_data$MATH,train_data$READING)
print(paste("Correlation Coefficient:",math_reading_cor$estimate))
```



```
## [1] "Correlation Coefficient: 0.862202752155349"
```

```
#math and reading predictions & RMSE
```

```
math_reading_predictions <- predict(math_reading_model,newdata = test_data)
math_reading_RMSE <- sqrt(mean((math_reading_predictions - test_data$MATH)^2))
print(paste("The RMSE for Math vs Reading is ", math_reading_RMSE))
```

```
## [1] "The RMSE for Math vs Reading is 0.0938937945097898"
```

From the graphs (Math vs. Reading), (Reading vs. Writing), and (Math vs. Writing), we observe that all the slopes are positive. This suggests that with an increase in passing rates for one subject, the passing rates for the other will also tend to increase. The relationship between Writing and Math suggests a stronger correlation between the two subjects which is shown through the steep slope

#RMSE of 2010 Math Score

```
#math passing rates
```

```
#linear model to predict math from reading and writing data
```

```
linmodel_math <- lm(MATH ~ READING + WRITING, data=train_data)
```

```
#summarizing the linear model
```

```
summary(linmodel_math)
```

```
##
```

```
## Call:
```

```
## lm(formula = MATH ~ READING + WRITING, data = train_data)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
## -0.254281 -0.047286  0.000958  0.060759  0.228055
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.12481    0.03713  -3.361 0.001049 **
## READING      0.25418    0.15154   1.677 0.096159 .
## WRITING      0.54603    0.14198   3.846 0.000196 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Residual standard error: 0.08203 on 117 degrees of freedom
```

```
## Multiple R-squared:  0.7722, Adjusted R-squared:  0.7683
```

```
## F-statistic: 198.3 on 2 and 117 DF, p-value: < 2.2e-16
```

```
#predicting the 2010 math passing rates using the model
```

```
test_data$predicted_MATH <- predict(linmodel_math,newdata = test_data)
```

```
#calculating the math RMSE
```

```
math_rmse <- sqrt(mean((test_data$MATH - test_data$predicted_MATH)^2))
cat("The RMSE for predicting the math passing rates in 2010:", math_rmse, "\n")
```

```
## The RMSE for predicting the math passing rates in 2010: 0.0915186
```

RMSE for predicting Math passing rates in 2010: 0.09905299

Summary of RMSE for Reading, Writing, and Math CSAP Data for Colorado 9th Graders:

```
## The RMSE for Reading vs Writing: 0.06858903
```

```
## The RMSE for Math vs Writing: 0.09572715
## The RMSE for Math vs Reading: 0.09389379
## The RMSE to predicting the Math passing rates based on reading and writing in 2010: 0.0915186
## The RMSE to predicting the Reading passing rates based on math and writing in 2010: 0.06622528
## The RMSE to predicting the Writing passing rates based on math and reading in 2010: 0.06935631
```

Advise

In this study, we can see a statistically significant increase in math passing rates from 2009 to 2010. Using bootstrap analysis, we estimated the confidence interval for the mean difference in math scores. Because the confidence interval did not include 0, this shows strong statistical evidence that the true mean increase in math scores is greater than 0. While there was no significant change in reading scores, there was a significant decline in writing scores within the same period. These findings can be used to inform targeted strategies to improve student performance. The association plots between the subjects showed strong positive correlations, as indicated by the upward slopes in the scatter plots. This would imply that the higher the passing rate in one subject, the higher the passing rates in the other subjects. Statistically, this would mean that schools that tend to have high passing rates in reading and writing also tend to have high math passing rates, which would allow for reasonable predictions of math scores based on the other subjects. However, it should be taken into consideration that this analysis doesn't account for outside factors or confounding variables such as teaching methods, student behavior, school curriculum, or classroom environment that affects the results we observed.

Answer

The analysis shows a strong statistical relationship between a school's passing rates in math, reading, and writing. This suggests that if a school performs well in one subject, it is likely to have high passing rates in the other subjects as well. By using reading and writing scores, we can make reasonably accurate predictions for math scores. The model's error, indicated by the Root Mean Squared Error (RMSE) of 0.92, demonstrates that the predictions are reliable and offer a good estimate of math scores. The confidence intervals also confirm an increase in math passing rates over time. In contrast, reading scores showed no significant change, while writing scores experienced a slight decrease. There is a strong association between math passing rates and both reading and writing passing rates. The RMSE values further supports the accuracy of the model's predictions for math scores based on reading and writing, indicating that the model provides a good approximation of math performance in schools.